

Vane: A Dual-Layer LLM Agent Monitoring Narrative Drift

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Abstract

We present **Vane**, an automated open-source AI agent tracking narrative drift on social media. Vane uses a dual-layer LLM pipeline: a lightweight model (*Mistral-Small*) handles structured daily context compression, while a large model (*GPT-OSS-120B*) performs multi-day temporal reasoning and comparative analysis. We test the system on Bluesky narratives of OpenAI, Anthropic, and Google DeepMind and find that it works well. The full code is available on GitHub (Hazelime, 2026).

1 Introduction

Traditional sentiment classification treats social media as static data points, failing to capture the fluid ways public opinion shifts during crises and other significant events (Farr et al., 2026). Our proposed solution, Vane, addresses this limitation by actively mapping longitudinal data streams to track how public narratives drift and evolve in real time.

In this report, we exemplify Vane’s capabilities by monitoring public opinion of OpenAI, Anthropic, and Google DeepMind on the social media platform Bluesky (Bluesky Social, PBC, 2026). While competitors like X and Reddit wall off their data behind restrictive APIs, Bluesky offers an open window into real-time public opinion. By harnessing this data stream, an automated AI agent can consistently detect early, subtle, and abstract shifts of narratives that may slip past human analysts and conventional tracking tools.

The full code is available on GitHub along with further instructions (Hazelime, 2026). A YouTube video also shows how to use the CLI subcommands `ingest`, `summarize`, `daily`, and `report` on a local installation (Bennevall, 2026).

2 Ingestion

Once a day, Vane accesses the Bluesky API (`app.bsky.feed.searchPosts`) to pull the top 100 posts mentioning OpenAI, ChatGPT, Anthropic, Claude, (Google) DeepMind, and Gemini, respectively. (Since "Gemini" is a polysemous term, such posts are discarded unless they contain context words such as `ai` or `llm`.) Retrieved posts are then scored using the following engagement formula:

$$S_{\text{eng}} = 1 \cdot L + 3 \cdot R_p + 4 \cdot R_y + 5 \cdot Q \quad (1)$$

Likes (L), reposts (R_p), replies (R_y), and quotes (Q) are weighted according to their relative importance.

Following deduplication, Vane performs thread expansion on the top 50 posts per company by fetching up to 3 direct replies via `app.bsky.feed.getPostThread`. This results in a maximum of 156 API ingestion calls daily.

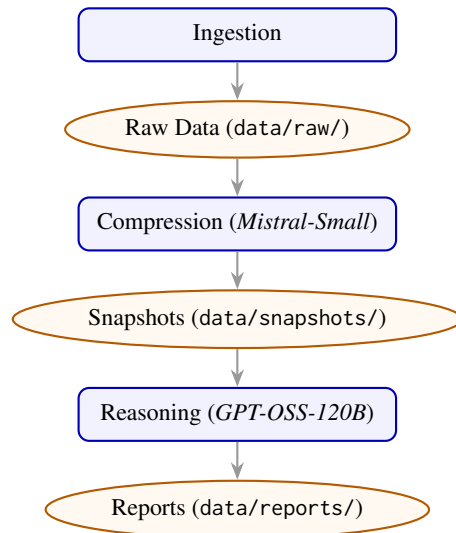


Figure 1: Vane data pipeline overview.

3 Dual-Layer LLM Inference

Vane employs two LLMs for inference: one lightweight model for daily data compression, and one more advanced model for multi-day temporal reasoning (Figure 1).

3.1 Daily Compression Layer (Mistral-Small)

Every day following the scheduled ingestion, Mistral-Small-3.2-24B-Instruct-2506 converts the noisy Bluesky data into a structured JSON snapshot for each company. These snapshots capture 1) dominant themes, 2) dominant sentiments, 3) notable events, 4) cross-company comparisons, 5) unusually engaging posts, and 6) emerging vocabulary signals.

3.2 Temporal Reasoning Layer (GPT-OSS-120B)

Every third day or when prompted manually, Vane passes a rolling 9-day window of snapshots to openai/gpt-oss-120b for trend analysis. The reasoning is guided by an explicit five-stage prompt:

1. **Baselines:** Establish the current baseline narrative for each company.
2. **Historical comparisons:** Compare the baselines with the previous 9 days.
3. **Cross-company contrasts:** Contrast the narratives of the three companies against each other.
4. **Causal links:** Identify likely drivers, such as significant events and vocabulary changes.
5. **Confidence:** Assign confidence to analyses. ("High confidence" is restricted to trends appearing across at least 3 consecutive snapshots.)

4 Results and Discussion

For demonstration purposes, we let Vane monitor the Bluesky narratives surrounding OpenAI, Anthropic, and Google DeepMind during May 14-19, 2026. Figure 2 shows two excerpts from the final report.

Vane has clearly picked up on news and comments about the Elon Musk vs. OpenAI trial and, furthermore, noticed that questions remain even after the

court ruling (BBC News, 2026; University of Western Australia, 2026), demonstrating temporal awareness and an eye for subtle details. The second excerpt also shows that Vane is able to identify both common themes and dissimilarities between narratives.

"The conversation around OpenAI has moved from a strongly negative focus on legal accountability and corporate misconduct (05-14) to a mixed tone after a jury ruled against Elon Musk (05-19). While the legal victory softens criticism, persistent worries about AI safety, data privacy, and regulatory capture keep the overall sentiment cautious."

"Legal battles dominate OpenAI and Anthropic, while DeepMind's discourse is driven by internal labor actions and product reliability."

Figure 2: Report excerpts summarizing changing public sentiment surrounding OpenAI (top) and theme similarities and differences between company narratives (bottom).

5 Conclusion

Vane proves that a dual-layer LLM architecture powered by decentralized social media endpoints offers a rigorous solution for long-term narrative trend analysis. While in this report it was demonstrated specifically on corporate narratives on Bluesky, the same architecture can easily be adapted to other contexts. Like a weather vane indicates the direction of the wind, Vane captures the drift of public opinion.

6 Reflection on AI Collaboration

Encouraged by optimistic LLMs, I had in fact planned to test Vane on subreddits instead. Begrudgingly late into the process, however, I found out that Reddit had recently closed off its API, forcing me to pivot to Bluesky. This taught me to verify the soundness of each step in a pipeline before building on it: a house is only as strong as its foundation.

Secondly, working with real-life money (via Berget) made this assignment feel even more high-stakes than the last assignment, where the main constraint was the weekly token quotas. Lacking experience with LLM-to-work "exchange rates," I spent many, many hours trying to perfect my initial prompt. Now, however, I can better appreciate how much €1 actually is (I only used €0.2 in one week!).

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